



Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

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Project Title

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1. Problem Statement

Ground Glass Opacities (GGOs) are faint cloudy areas seen in lung CT scans. They can be early signs of diseases such as pneumonia, lung infections, or even cancer. Finding these areas by hand is difficult and takes a lot of time because GGOs have low contrast and spread out gradually in the lungs. Different doctors may also interpret them differently, which can affect diagnosis accuracy.

Although deep learning methods are used in medical imaging, many existing systems mainly focus on identifying whether a disease is present, rather than clearly showing where GGOs are located in the CT scan.

This project proposes an image processing and computer vision based tool that automatically highlights GGO regions in lung CT scans. The system processes raw CT images and converts them into clear and easy to understand visual outputs. This tool is designed to support radiologists by improving the accuracy of interpretation and helping with early detection and monitoring of lung abnormalities.

2. Objectives

- Improve the quality of the chest CT scans by removing the noise, normalizing the image intensity values, and converting the data into a format suitable for processing.
- Segmentation of the lung regions from the CT volumes using basic intensity based methods, and morphological image processing techniques
- Show the detected GGO areas on the CT images by overlaying them visually or using pseudo coloring enhancement to highlight their visibility.
- Test the efficiency of the proposed image processing workflow on a publicly available chest CT dataset.
- Evaluate the effectiveness of the proposed image processing pipeline using a publicly available chest CT dataset.
- Design and show a simple prototype application which focuses on clear image visualization and the ease of interpretation of the results.

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3. Introduction

Computed tomography has become indispensable for pulmonary diagnostics. It enables visualization of subtle parenchymal abnormalities including ground glass opacities. These semi transparent lesions indicate various pathologies from inflammatory processes to neoplastic changes. Manual GGO identification demands considerable radiologist time while introducing inter observer variability. This review examines established preprocessing methodologies, lung parenchyma isolation techniques, GGO detection algorithms, and vessel suppression methods. It ultimately identifies research gaps necessitating interactive visualization tools.

4. Study 1: Hounsfield Unit Based Lung Segmentation

Initial automated lung isolation relied upon intensity thresholding exploiting the distinct radiodensity of air filled pulmonary tissue. Foundational work established that attenuation values measured in Hounsfield units provide well defined ranges for different tissue components on CT images [1]. This principle enabled straightforward binary classification separating lung parenchyma from surrounding thoracic structures. However, pathologic areas within lungs frequently exhibit altered densities, complicating simple threshold based approaches.

A fully automatic three dimensional segmentation method combined gray level thresholding with dynamic programming and morphological operations to refine initial binary masks [2]. The dynamic programming component optimized boundary placement where intensity gradients were weak or ambiguous. Morphological refinement addressed typical artifacts including incomplete boundary closure and spurious small regions. This methodology established performance benchmarks for subsequent comparative studies.

Practical implementation frameworks have emerged providing accessible workflows. One tutorial demonstrated lung segmentation using Hounsfield ranges between -1000 and -500, implementing morphological closing operations for each lung component separately to prevent artificial fusion [3]. The separate processing proved essential for maintaining anatomical accuracy, particularly near the mediastinum where left and right lungs approximate closely. These workflows typically employ Python based libraries offering rapid prototyping capabilities.

5. Study 2: Advanced Segmentation for Diseased Lungs

Pathological conditions substantially complicate automated segmentation since disease processes alter normal tissue characteristics. Research addressing severe interstitial lung disease employed CT value thresholding combined with texture feature images derived from co occurrence matrices [4]. The texture analysis captured spatial patterns beyond simple intensity values, enabling differentiation between diseased lung parenchyma and

non pulmonary structures. This approach achieved 82% overlap rates with manual annotations despite challenging anatomical variations.

Hybrid methodologies integrating fuzzy clustering with morphological processing have demonstrated improved robustness. One investigation used fuzzy c-means clustering combined with hole filling, dilation, and morphological reconstruction specifically targeting juxta-pleural nodules that merge with chest walls [5]. The fuzzy clustering accommodated gradual transitions rather than imposing hard boundaries, better matching biological reality. Morphological reconstruction selectively filled gaps while preserving anatomically meaningful boundaries. These techniques proved particularly valuable when consolidative or fibrotic changes obscured normal lung tissue boundaries.

6. Study 3: Ground Glass Opacity Detection Algorithms

Identifying GGO regions within segmented lungs requires distinguishing subtle density increases from normal parenchymal attenuation. Probabilistic spatial modeling through Markov random field based algorithms achieved 86.94% sensitivity and 94.33% specificity by incorporating contextual relationships between neighboring voxels [6]. The Markov framework explicitly modeled the characteristic diffuse spatial distribution of ground glass abnormalities. Additionally, this work implemented vessel removal using morphological operations targeting elongated line like structures, recognizing vascular anatomy as a primary false positive source.

Statistical characterization approaches offer complementary perspectives on GGO identification. Research utilizing morphological reconstruction for initial segmentation subsequently performed statistical analysis including mean, standard deviation, variance, entropy, skewness, and kurtosis calculations [7]. These first order statistics quantified intensity distributions within candidate regions, enabling probabilistic classification. Entropy measures proved particularly informative, capturing the heterogeneous texture characteristic of many ground glass patterns. The statistical framework facilitated confidence scoring rather than binary classification.

Texture based methods emphasizing local pattern analysis have shown particular promise for nodular GGOs. An investigation developed a classifier using boosted k-nearest neighbor with nonparametric density estimates, followed by automatic segmentation through texture likelihood map analysis [8]. This approach successfully detected all ten GGO nodules in the test cohort with minimal false positives. The texture likelihood mapping technique analyzed local neighborhood patterns, identifying regions exhibiting characteristic GGO heterogeneity. The integration of detection and segmentation stages created a cohesive processing pipeline.

Recent clinical demands, particularly from COVID-19 diagnostics, have motivated efficiency focused innovations. An attention mechanism threshold method addressed GGO segmentation challenges related to variable features and low intensity contrast, improving Dice similarity by 8.9% while requiring only 0.09% of deep learning computational requirements [9]. The attention mechanism selectively emphasized diagnostically rele-

vant regions while suppressing background areas. This computational efficiency enables deployment on standard clinical workstations without specialized hardware.

Three dimensional geometric analysis techniques have introduced novel characterization approaches. PointNet++ architecture applied to volumetric GGO analysis achieved 98% evaluation accuracy and 92% intersection over union, while quantifying morphologies using Minkowski tensors [10]. These mathematical descriptors captured shape anisotropy and complexity beyond simple volume measurements. The geometric characterization enabled differentiation between GGO subtypes based on three dimensional morphological features, potentially correlating with underlying pathophysiology.

7. Study 4: Pulmonary Vessel Segmentation and Suppression

Vascular structures present a significant challenge for GGO detection since vessels and ground glass opacities share overlapping Hounsfield value ranges. Multiscale filtering techniques analyzing Hessian matrix eigenvalues have become standard for vessel enhancement. A comprehensive three dimensional multiscale approach enhanced vascular structures while suppressing non vessel elements such as lymphoid tissue, achieving 96.2-97.0% accuracy in vessel tree segmentation [11]. The multiscale framework accommodated the wide diameter range of pulmonary vasculature from major vessels to peripheral branches. Eigenvalue analysis identified tubular structures based on local geometric properties rather than intensity alone.

Dynamic contour algorithms have addressed the extraction of connected tubular networks. These methods employed threshold segmentation followed by morphological principles and elliptical fitting based on the ratio of long and short axes characteristic of vessel cross sections [12]. Edge contour tracking maintained continuity along vessel pathways despite local intensity variations. The geometric constraints ensured detected structures exhibited expected tubular morphology, filtering spurious detections.

Comprehensive vessel characterization frameworks have provided methodologies applicable to removal contexts. Studies examining vessel tortuosity implemented automatic segmentation using gray level thresholding followed by morphological closing operations, with vessel enhancement filters using Hessian matrix eigenvectors to detect tubular structures [13]. The enhancement filters selectively amplified tubular patterns across varying contrast conditions. Distance transform techniques subsequently enabled quantification of vessel proximity, informing region specific processing strategies.

8. Gap Analysis

Despite considerable research progress, several critical limitations persist in current GGO detection methodologies. First, most published systems function as automated processing pipelines lacking interactive user control during intermediate stages. Radiologists cannot adjust parameters based on case specific anatomical variations or pathology characteristics, reducing clinical workflow integration. Second, vessel removal techniques typically

apply uniform parameters across entire lung volumes, ignoring regional differences in vascular density from central hilar regions to peripheral parenchyma. This uniform approach generates either incomplete vessel suppression centrally or excessive parenchymal erosion peripherally.

Third, validation studies predominantly report aggregate performance metrics without analyzing failure modes or edge cases where automated methods perform poorly. Understanding specific scenarios causing algorithmic failures would inform targeted improvements. Fourth, few systems provide volumetric quantification with slice by slice navigation enabling correlation between three dimensional extent and clinical severity. Fifth, transparency adjustment for overlay visualization remains underexplored, though this feature significantly impacts radiologist interpretation by enabling rapid comparison between original and processed images.

Finally, computational efficiency analyses rarely consider complete preprocessing pipelines including DICOM loading, windowing, segmentation, and visualization. Individual algorithmic components may demonstrate efficiency, yet cumulative processing time determines clinical viability. These gaps justify developing interactive visualization tools offering adjustable parameters, regional processing strategies, comprehensive validation metrics, volumetric navigation, overlay transparency control, and complete pipeline optimization. Such tools would bridge the gap between automated processing capabilities and practical clinical deployment requirements.

9. Methodology

(Minimum word limit: 1000 words) Describe the approach, tools, and algorithms you will use. Mention the main image processing or computer vision techniques (e.g., segmentation, feature extraction, classification) and the software frameworks (e.g., OpenCV, MATLAB, TensorFlow).

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10. Dataset / Data Collection

(Maximum word limit: 100 words) Provide details of the data its source, number of images, format, and characteristics. If you plan to collect your own data, explain how you will ensure data quality, diversity, and ethical use.

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11. Expected Outcomes

(Maximum word limit: 100 words) Summarize the expected outputs (eq: demo, code files, presentation) and performance metrics (e.g., accuracy, clarity, speed). Explain how your results will demonstrate the effectiveness of your image processing or computer vision approach. Write the outcomes in point form.

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12. Percentage of Image Processing & Computer Vision Involvement

(Approximately 50–100 words) State clearly how your project satisfies the requirement of at least 65% image processing and computer vision components. For example, describe which parts involve preprocessing, segmentation, or model development etc. And if you are planning to use any tools may be you can mention.

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A minimum of ten (10) references must be included in the References section.